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FlintNDE: a FLINT-based package for high-precision numerical solution of matrix differential equations

Jiaqi Chen[a,b]

[a] *Beijing Key Laboratory of Optical Detection Technology for Oil and Gas, China University of Petroleum-Beijing, Beijing 102249, China*

[b] *Basic Research Center for Energy Interdisciplinary, College of Science, China University of Petroleum-Beijing, Beijing 102249, China*

E-mail: jiaqichen@cup.edu.cn

Abstract: First-order matrix differential equations with rational coefficient matrices form the shared numerical core of three lines of calculation: dimensionally regularized Feynman integrals in flat-spacetime quantum field theory, cosmological correlators in de Sitter space obtained through the same integration-by-parts and differential-equation strategy, and radial connection problems in black-hole perturbation theory. We present `FlintNDE`, a Python package that solves such systems along paths in the complex plane to user-specified high precision. It is built on the FLINT library and performs all arithmetic in midpoint-radius ball arithmetic, so that every returned value carries a rigorous error estimate. Local solutions are generalized power series, that is, a power factor, powers of a logarithm and a Taylor remainder. Wherever the input is exact, the underlying structure (singularity locations, indicial spectrum, logarithm degrees) is certified in exact Gaussian-rational arithmetic. As one further capability, decoupled essential singularities, where the high-order pole part splits into scalar sectors, are handled with certified exponential-polynomial phases. The package plans paths adaptively around the certified singularities, separates planning from execution, records reusable save points, reconstructs Laurent series in an analytic regulator following the sampling procedure of AMFlow, and evaluates batches of regulator values in parallel. A Wolfram Language interface exposes the same functionality. We demonstrate all eight shipped examples, including the two-ended connection problem for the $\ell = 2$ Schwarzschild quasinormal mode, and verify the results against independent references.

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1 Introduction

High-precision numerical evaluation is now a standard step in three families of physics calculations. In flat-spacetime quantum field theory, multi-loop Feynman integrals enter cross sections for collider processes and are reduced to master integrals by integration-by-parts (IBP) relations [1], whose induced differential equations (DE) [2–5], often brought to the canonical d log form [6], are then solved from boundary data. In de Sitter (dS) space, late-time correlation functions connect the dynamics of fields during inflation to the statistical properties of primordial fluctuations [7–9], and the same IBP-to-DE strategy has recently been extended to their time integrals [10–12]. In black-hole physics, the linear perturbations of the Schwarzschild geometry separate into radial Regge–Wheeler and Zerilli equations [13, 14], and their quasinormal modes (QNM)—the ringdown signals observed in gravitational-wave detections [15]—are defined by a two-ended boundary-value problem [16, 17]. All three settings reduce, after symbolic preparation, to the same numerical object: a first-order system $\mathrm{d}Y/\mathrm{d}z = A(z)\,Y$ with rational or analytic coefficient matrix, to be transported reliably between points of the complex plane.

A mature toolbox exists on the symbolic side. FIRE [18, 19], Kira [20–22] and AmpRed [23] generate IBP reductions; expansion by regions [24, 25] and dimensional recurrence [26] provide complementary asymptotic information. On the numerical side, DiffExp [27] transports solutions along piecewise-linear paths in terms of one-dimensional series expansions, and the auxiliary mass flow method of AMFlow [28–30] solves the DE at a sequence of auxiliary mass values and reconstructs the Laurent series in the dimensional regulator. Both are Mathematica packages. The recent AMFlow 2.0 release additionally provides a high-performance differential-equation solver written in C++, connected to Mathematica through WSTP and also available as a standalone, configuration-driven command-line executable that reads the differential-equation matrix and the boundary values from plain input files [31]. Along the path, the local series that both methods transport are generalized power series—a power factor, powers of a logarithm, and a Taylor remainder—as introduced for Feynman-integral differential equations in ref. [32].

`FlintNDE` is a plain, fast numerical solver for exactly this class of systems, implemented in Python on top of the FLINT library [33, 34]. It takes an already-reduced matrix differential equation together with certified boundary data and returns values, local series and Laurent coefficients to user-specified precision; it does not generate IBP relations and does not manage integral families. Where the `desolver` executable packages the AMFlow differential-equation module for users who want it without the rest of the pipeline, `FlintNDE` additionally provides, for the same class of systems, explicit path planning with singularity avoidance and jumps, coverage of dense sets of requested points by shared series coefficients (section 3.5), and certified exponential-polynomial phases at decoupled essential singularities (section 3.3). Its design choices are the following.

1. All floating-point arithmetic uses Arb ball numbers [33]: every returned component carries a radius, and the default working precision is 200 decimal digits. Interval results, not point estimates, decide whether a requested relative error has been met.

2. Whenever the input matrix is exact over Gaussian rationals, the structural decisions are made in exact arithmetic: singularities are discovered from the matrix entries, the matrix indicial equation is solved over $\mathbb{Q}(\mathrm{i})$, and logarithm degrees follow from Jordan structure and resonance conditions. Where a certified structure is required but unavailable, the package reports failure instead of silently degrading.

3. Local solutions are the generalized power series of ref. [32]. As one additional capability, decoupled essential singularities, whose high-order pole part splits into scalar sectors, are handled by exact exponential-polynomial phases, and a start-only formal asymptotic branch exists for second-order poles with simple distinct leading spectrum; no systematic treatment of general high-order poles is attempted.

4. Path planning is separated from execution. Plans serialize with their precision, avoid certified singularities by default, and can be executed only at or below the planning precision. Explicit singularity jumps and strictly user-supplied node chains are available options.

5. Laurent reconstruction in an analytic regulator reuses the complete numerical transport at exact rational regulator values and follows the empirical sampling plan of AMFlow 2.0 [31], with an explicit candidate-pool semantics and independent validation points.

6. The same functionality is exposed through a Wolfram Language interface, and batches of fixed regulator values can be evaluated in a bounded process pool.

The package is used as the numerical backend of the `MadStree` package for dS tree correlators, where it transports d log-form systems and reconstructs regulator series, and independently for black-hole radial problems. Sections 2 and 3–4 specify the mathematical object, the transport engine and the regulator reconstruction. Section 5 describes usage from Python and from Wolfram Language. Section 6 presents the eight shipped examples, including the Schwarzschild QNM connection problem, and section 7 reports verification against independent references.

2 The problem and the interface contract

2.1 Systems and input formats

We consider column-vector systems

$$\frac{\mathrm{d}}{\mathrm{d}z}Y(z) = A(z)\,Y(z), \qquad A(z) \in \mathbb{C}^{N\times N}, \tag{2.1}$$

where $A(z)$ is either an exact matrix of rational functions over $\mathbb{Q}(\mathrm{i})$ or an arbitrary analytic callable. The exact class is entered as coefficient records with constant term first,

$$A_{ij}(z) = \frac{p_{ij}(z)}{q_{ij}(z)}, \qquad p_{ij},\, q_{ij} \in \mathbb{Q}(\mathrm{i})[z], \tag{2.2}$$

and every coefficient accepts integers, fractions, decimal strings or `"a+b*I"` records. Floating-point input is rejected on the exact route: raising the working precision cannot recover input digits that were never there.

For exact input, finite singularities, their pole orders and the classification of infinity are discovered internally from the denominators; the user supplies neither pole positions nor residues. The solver also recognizes the special structure

$$A(x) = P(x) + \sum_{j} \frac{R_j}{x - p_j}, \tag{2.3}$$

where $P(x)$ is a matrix polynomial of arbitrary finite degree; an explicitly known decomposition of this form may be supplied directly, which is an optimization, not a requirement. General rational matrices and analytic callables use the generic route, and pure polynomial systems form the legal special case of an empty pole set. Section 3.5 describes when the transport verifies identity (2.3) internally and switches to a fast recurrence route.

2.2 Local solutions as generalized power series

At an ordinary point c, write $s = z - c$ and expand $A(c+s) = \sum_{m\ge 0} A_m(c) s^m$. The solution coefficients obey the triangular recurrence

$$Y_{n+1}(c) = \frac{1}{n+1} \sum_{m=0}^{n} A_m(c)\, Y_{n-m}(c). \tag{2.4}$$

The matrix Taylor coefficients $A_m(c)$ are recovered numerically by Cauchy sampling—evaluations on a circle followed by a discrete Fourier projection—which requires only analyticity inside the sampling disc.

At a regular singular point, a Fuchsian system $\mathrm{d}Y/\mathrm{d}z = (R/z + \sum_{m\ge 0} A_m z^m)Y$ has local solutions

$$Y = z^{\rho} \sum_{n\ge 0} \sum_{q=0}^{q_{\max}} v_{n,q}\, z^n (\log z)^q, \tag{2.5}$$

where the exponents ρ solve the matrix indicial equation

$$\det(\rho \mathbb{I} - R) = 0, \tag{2.6}$$

and the coefficients obey, at fixed (n, q),

$$\big((\rho+n)\mathbb{I} - R\big) v_{n,q} + (q+1)\, v_{n,q+1} = \sum_{m=0}^{n-1} A_m\, v_{n-1-m,q}. \tag{2.7}$$

Two distinct mechanisms produce logarithms. A Jordan block of length ν in R gives $z^{\rho} \sum_{q<\nu} (N^q/q!)(\log z)^q$ with $N = R - \rho\mathbb{I}$. Even for a diagonalizable residue, equation (2.7)

becomes singular when $\rho + n$ is another indicial root: a pure-power recurrence survives only if the spectral projector compatibility $P_{\rho+n} b_n = 0$ holds, and the defect otherwise migrates into a logarithmic coefficient. The package decides both mechanisms in exact rational arithmetic.

Boundary data at a singular start are therefore specified as leading branches

$$\{a,\ b,\ C\}\ :\qquad Y(z) \sim z^a (\log z)^b\, C + \cdots, \tag{2.8}$$

with a and the components of C exact over $\mathbb{Q}(\mathrm{i})$, b a non-negative integer and C of the system dimension; several records may be superposed. For example, the branch $3x^2$ of $\mathrm{d}y/\mathrm{d}x = 2y/x$ is entered as $\{a{=}2,\ b{=}0,\ C{=}[3]\}$. The vector C must lie in the certified eigenspace (or, for a Jordan branch, satisfy $NC = 0$ and lie in the image of N^b); lower logarithmic degrees of a Jordan chain are completed automatically from $\exp(N \log z)$.

At a decoupled essential singularity, all Laurent matrices of pole order two and higher are simultaneously diagonal in the eigenbasis of the leading one, and lower-order terms do not couple sectors with different exponential signatures. The scalar high-order part $\sum_{m\geq 2} c_m z^{-m}\mathbb{I}$ is then removed by

$$Y = e^{\Phi(z)}\, U, \qquad \Phi(z) = -\sum_{m=2}^{p} \frac{c_m}{m-1}\, z^{-(m-1)}, \tag{2.9}$$

after which the residual system must have pole order at most one and the power–log machinery of equation (2.7) applies to U. The corresponding boundary records extend to

$$\{\phi,\ a,\ b,\ C\}\ :\qquad Y(z) \sim e^{\phi(z)}\, z^a (\log z)^b\, C + \cdots, \tag{2.10}$$

where the phase ϕ lists the coefficients of the negative powers of the local variable, with exact values derived from the differential equation and verified entry by entry; for $\mathrm{d}y/\mathrm{d}x = y/x^2$, whose solutions are $C\, e^{-1/x}$, it is the single term $\phi = \{-1 \mapsto -1\}$. This extends the generalized power series of ref. [32] beyond the logarithmic structure for this certified class of singularities.

2.3 Scope

The package consumes matrix differential equations and boundary data; it does not generate IBP relations, does not read integral families, master orderings or normalizations, and does not infer the Laurent support of a regulator expansion from numerical samples—the lowest Laurent power must be certified upstream. Structures that would require ramified formal gauges, algebraic field extensions beyond $\mathbb{Q}(\mathrm{i})$, general formal block decoupling or Stokes connection matrices are not implemented; whenever the certified route to a requested local basis does not exist, the package raises an exception rather than returning an uncertified value. Section 3.3 delimits the irregular case precisely.

3 Core function I: numerical transport

3.1 Overview

A calculation proceeds in five stages. First, the exact system is analyzed and a singularity inventory is produced: locations, pole orders, and the classification of each point as ordinary,

regular singular or irregular, including the point at infinity. Second, local solution bases are constructed at the requested endpoints. Third, a path is planned in the complex plane, with planning separated from execution (section 3.4). Fourth, the solution is transported segment by segment: each segment carries the Taylor recurrence of equation (2.4) inside a Cauchy disc that contains no singularity. Fifth, results are written as save points together with a summary, and a two-chain error estimate is returned (sections 3.4 and 3.6). Figure 1 summarizes the pipeline.

3.2 Exact structure certification and the numerical route

For exact input, the indicial roots, Jordan nilpotency, spectral projectors and every active resonance gate are determined by FLINT rational matrices and stored in a manifest; FLINT then evaluates equation (2.7) at the requested precision. An exact failure never falls back to a threshold decision.

When the original input is floating-point data, an explicit numerical route exists. It classifies the structure with a relative cutoff scaled by the system dimension and by the reliable decimal precision of the input—Python binary64 input is capped at 15 digits regardless of the working precision, and higher-precision input must declare its reliable precision. Every numerical classification emits a warning and a manifest recording the matrix scale, both cutoffs, and the largest zeroed and smallest retained quantities; ambiguous mixed-root defective systems fail closed. The exact cutoff formula is given in appendix B.

3.3 High-order poles

A pole of order $p \geq 2$ in some matrix entry triggers a three-route dispatch. First, an exact Lee–Moser projector balance is attempted: from the top two Laurent matrices a projector Q with $Q^2 = Q$ is constructed and the gauge transformation

$$I = \left(1 - Q + \frac{Q}{z}\right) J \tag{3.1}$$

is applied; each round must lower the pole order strictly, and the complete transformed system is verified by an exact round trip. Success reduces the point to the power–log case, with the boundary transformed back to the original basis.

If a high-order pole remains, the second route applies at decoupled essential singularities as defined in section 2.2: the solution is the exponential factor of equation (2.9) times a convergent power–log series. The third route applies when the pole order is exactly two and the leading matrix has simple distinct eigenvalues k_j over $\mathbb{Q}(\mathrm{i})$: for each eigendirection it constructs the formal branch

$$Y = e^{-k/z}\, z^{\rho} \sum_{n=0}^{N_{\mathrm{form}}} v_n z^n, \qquad \rho = \frac{\ell^T A_{-1} v_0}{\ell^T v_0}, \tag{3.2}$$

with an exact recurrence for the v_n. This is a sectorial asymptotic series, not a convergent Frobenius disk; it may only initialize a path start, and the package reports a five-order

block ratio and relative refinement

$$r_5 = \frac{\left\|\sum_{j=1}^{5} T_{N+j}\right\|_\infty}{\left\|\sum_{j=0}^{4} T_{N-j}\right\|_\infty}, \qquad \delta_5 = \frac{\|S_{N+5} - S_N\|_\infty}{\|S_{N+5}\|_\infty} \tag{3.3}$$

with $T_n = v_n z^n$, together with the least-term degree actually reached. The first ordinary matching point is proposed from the exponential root gaps, $|z| = \min_i \Delta_i/(qN)$ with $\Delta_i = \min_{j\neq i} |k_i - k_j|$ and default safety factor $q = 3$. All three routes fail closed on unsupported structure—defective leading matrices, spectra outside $\mathbb{Q}(\mathrm{i})$, residual high-order poles after the scalar extraction—with messages naming the missing reduction.

3.4 Paths, singularities and transport

Endpoints are named points; the literal string `"inf"` denotes infinity, which is mapped to the zero of the inverted variable $\mathrm{sinv} = 1/z$ with transformed matrix $-\mathrm{sinv}^{-2}A(1/\mathrm{sinv})$. Paths are generated adaptively: each step is bounded by a configurable fraction (default 0.45) of the controlling convergence radius, which is the distance to the nearest singularity whenever one exists. Planning and execution are strictly separated: `FlintNDE` (Python entry `plan_transport_path`) produces a complete serialized plan—including node geometry with ball midpoints and radii and the planning precision—and execution (`transport_planned_path_refined`) restores and executes it without replanning; on the Wolfram route, an execution requesting higher precision than the plan is rejected and must be replanned.

By default the planner avoids certified singularities and returns structured diagnostics when a straight segment hits one. The user may supply ordinary detour points, or enable explicit singularity jumps, in which case the internal singularity is represented by a local basis bridge between ordinary matching points rotated off the segment; the selected multivalued branch is equivalent to some detour path and the winding numbers are reported, since the responsibility for the branch choice stays with the user. A strict user-node chain (`direct_user_point_path`) disables planning entirely: no points are inserted or removed, and hitting a singularity fails. A separate interface evaluates a user-requested point that coincides with a finite singularity: removable components return values, while genuine pole and logarithmically divergent components return the classification text `Infinity` together with per-component orders.

Requested output points may carry a `"save"` tag. Save points are written immediately as they are reached—ordinary points as vectors, singular points as reusable $\{a, b, C\}$ or $\{\phi, a, b, C\}$ boundary records—and a failure later in the path leaves the completed files intact and suppresses the summary.

3.5 Efficiency

Four features keep the solver fast without affecting the certified semantics.

1. *Certified simple-pole recurrence.* Once equation (2.3) is verified exactly—by partial-fraction decomposition of every entry, reconstruction of the right-hand side in $\mathbb{Q}(\mathrm{i})(x)$,

and independent ball-arithmetic probes at four sample points—segment transport no longer performs Cauchy sampling: the polynomial part is shifted by a binomial formula and each pole contributes a geometric-series state $u_j[n] = (c[n] - u_j[n-1])/(c - p_j)$, one step per Taylor coefficient, with residues read directly from the Laurent expansion.

2. *Lookahead node selection and fast multipoint evaluation.* While planning, the walker scans the remaining requested points in path order and places the next node at the farthest requested point that, together with every intermediate point, still lies within $0.9\,\kappa R$ of the current node, where R is the distance to the nearest certified singularity and $\kappa = \min(\kappa_{\text{user}}, 1/4)$ caps the step fraction. The points it steps over are not transported again: they become interior evaluation samples that share the node's Taylor coefficients, so dense sets of requested points are covered by far fewer transport steps. Samples of one segment form a bucket; buckets of eight or more points are evaluated by the subproduct-tree algorithm of `acb_poly`, smaller ones by iterative evaluation. The algorithm actually used is recorded per bucket.

3. *Batched singular starts.* Several Frobenius boundary vectors of one system share the local basis and the per-segment Taylor matrices; only the column-wise error gates are computed separately.

4. *Process-level parallelism.* Because the FLINT precision context is process-global, batches of independent fixed-regulator tasks run in a bounded spawn pool (default twelve workers) instead of threads.

3.6 Error control

Every transport runs two chains: the primary chain at the production order produces all user-visible values, and the reference chain at a higher order serves only for certification; their relative difference at the final endpoint is compared against the requested target, in interval arithmetic, and evaluated singular targets are certified separately. Alternatively, an embedded certification mode reuses the leading prefix of a single higher-order solve and reports the per-segment truncation differences. For formal asymptotic branches, the five-order diagnostics of equation (3.3) are reported. These residuals are error estimates, not rigorous bounds on the final error, and the returned status fields must be inspected.

4 Core function II: regulator-series reconstruction

4.1 Problem and certificate

Dimensionally regularized quantities take the form $F(\varepsilon) = \sum_{n=n_{\min}}^{n_{\max}} f_n \varepsilon^n + O(\varepsilon^{n_{\max}+1})$. Instead of truncating every intermediate expression in ε, the complete numerical problem is solved at several small exact rational values ε_j and the coefficients are reconstructed; this is the strategy of AMFlow [30]. A finite set of numerical solves cannot prove the Laurent support of a general callable, so the interface requires the integer $n_{\min}$ together with a strict upstream certificate (status, consistent integer, and method identification); missing

or inconsistent certificates fail before any solve. In the `MadStree` workflow the certificate is generated automatically from the symbolic boundary and d log-form DE, while `FlintNDE` itself remains independent of those data structures. Figure 2 summarizes the workflow; every regulator sample invokes the complete transport pipeline of figure 1.

4.2 Sampling plan

With $P = \max(0, -n_{\min})$ and $K = n_{\max} - n_{\min}$, the automatic plan follows the empirical configuration of AMFlow 2.0 [31] after the replacements $2L \to P$ and $L/2 \to P/4$ (the loop count L is specific to the Feynman-integral pole bound and is not a parameter of a general matrix system):

$$\begin{aligned} N_{\text{fit}} &= \max\left(\left\lceil \tfrac{5}{2}K + P \right\rceil, K + 3\right), & \alpha_\varepsilon &= \tfrac{P}{4} + \frac{g}{K+1}, \\ \varepsilon_0 &\simeq 10^{-\alpha_\varepsilon}, & \varepsilon_j &= \varepsilon_0\,(1 + j\,\delta), \quad j = 1, \ldots, N_{\text{fit}}, \\ p_0 &= \max(\lceil (N_{\text{fit}} + P)\alpha_\varepsilon \rceil, 30), & p_{\text{work}} &= \max(200, \lceil 2(1 + p_{\text{extra}})p_0 \rceil). \end{aligned} \tag{4.1}$$

where g is the digit goal, δ the relative sample spacing (default 0.01), and p_0 the transport-error goal in digits; the $K + 3$ branch includes the default `fit_extra_order` $= 2$. The transport uses production order $4p_0$ and reference order increased by $\lceil \max(50, p_0/5) \rceil$. The 200-digit floor is a `FlintNDE` default; explicit user values override automatic planning.

All N_{fit} production samples enter a single square Acb Vandermonde solve for the internal powers $n_{\min}, \ldots, n_{\min} + N_{\text{fit}} - 1$; only coefficients through the requested power $n_{\max}$ are returned, the extra powers absorbing truncation effects as in the AMFlow construction. No least-squares or pseudo-inverse fit is used.

4.3 Candidate pools, validation and complex sectors

An explicit `sample_points` sequence is an ordered production candidate pool: the first round consumes only the prefix required by the current square system, and a failed validation round consumes later candidates while reusing all previous solves. Once the pool is exhausted, the package never invents regulator values outside the user-approved range: it returns the current best coefficients with `precision_target_met=False` and the structured reason `candidate_pool_exhausted`. Independent validation points—by default on a scaled production grid, or explicit and disjoint—never enter the interpolation and are checked against both transport chains and the truncated series. Alternatively, `sample_angle_range` restricts automatic complex sampling to an open angular interval; at most three interior rays are selected, boundary rays are never used, and radii continue to follow equation (4.1). A scalar fitting layer (`fit_sampled_series`) reconstructs Laurent coefficients from samples the caller has already computed, for quantities such as Wronskians that are not a single transport endpoint.

5 Using the package

5.1 Installation and loading

The package requires Python ≥ 3.10 with `python-flint` ≥ 0.6 and `sympy` ≥ 1.12, and installs from a wheel, from source, or in editable mode; on first `import flintnde` it prints a

one-line citation reminder to standard error (suppressible through an environment variable, so that machine-readable standard output is never polluted). Importing establishes the default 200-digit global precision; `configure_working_precision(digits, guard)` resets it, with working bits $\lceil \text{digits} \cdot \log_2 10 \rceil + 32$. The Wolfram Language interface is loaded by adding the package root to `$Path` and calling `Needs["FlintNDE‘"]`; it launches the Python bridge with an argument-list `RunProcess`, writes bridge files under a short-token directory, and reports its own error labels (for example `PythonFlintUnavailable`, `BridgeLaunchFailed`, `RuntimePathTooLong`) when the runtime is misconfigured.

5.2 Python interface: basic commands

A calculation assembles an exact system, obtains the singularity inventory, plans a path, and transports:

```
from flintnde import *
layout = initialize_output_layout(__file__, run_name="run01")
system = RationalMatrixSystem(entries, variable_name="s")
inventory = analyze_singularities(system, output_layout=layout)
path = build_adaptive_path(system, NamedPoint("start", 2),
                           NamedPoint("target", 3),
                           max_step_over_radius=0.45)
result = transport_path_refined(
    system, column_vector([1, 0]), path,
    primary_order=24, reference_order=32,
    target_relative_error="1e-15")
# result["primary_snapshots"], result["relative_difference_inf"], ...
```

Entries are built with `rational_function`, whose numerator and denominator coefficients accept all exact formats of section 2.1; sums of rational functions are merged and cancelled exactly before denominators are collected. Singular starts use the boundary records of equations (2.8) and (2.10) through `frobenius_boundary` and `exponential_boundary`, which validate the exponents and vectors against the exact manifest before any transport. Save points are requested by appending the string `"save"` to a coordinate, and a capability pre-check (`build_adaptive_path_plan`) classifies all endpoints and internal singularities without numerical work, returning whether the requested continuation is certified.

5.3 Python interface: reconstruction and parallel evaluation

The minimal reconstruction call passes the complete numerical problem, the highest requested power, and the upstream certificate:

```
result = reconstruct_series_solution(
    DEmatrix=DEmatrix, boundary=boundary, path=path,
    maximum_power=2, leading_power=-1,
    leading_power_certificate={
        "status": "certified", "leading_power": -1,
        "method": "symbolic-boundary-and-DE"})
```

If the matrix, boundary or path depend on the regulator, they are passed as factories that receive exact rational regulator values, keeping every automatically generated sample exact. The result exposes `coefficient(power)`, `evaluate(ep)`, the production and validation values, the effective parameters, and per-round diagnostics. Independent fixed-regulator tasks run through `run_ep_tasks`, which starts at most `parallel_task_count` (default 12) workers and hands the next task to whichever worker finishes first.

5.4 Wolfram Language interface

The Wolfram interface mirrors the plan–execute separation of section 3.4:

```
Needs["FlintNDE`"];
system = FlintNDERationalSystem[{{1/(x - 1)}}, x];
plan   = FlintNDEPlanPath[system, 0, {1/2},
           "WorkingPrecisionDigits" -> 100,
           FlintNDE`SingularityMode -> "Avoid"];
result = FlintNDEExecutePath[system, {1}, plan,
           "PrimaryOrder" -> 40, "ReferenceOrder" -> 48,
           "TargetRelativeError" -> "1e-30"];
```

`FlintNDERationalSystem` accepts any exact rational matrix and discovers the singularities internally; `FlintNDEPartialFractionSystem` constructs an explicitly known decomposition of the form (2.3), which is an optimization and never a requirement. `FlintNDEEvaluateEpBatch` evaluates a list of fixed-regulator jobs with the option `ParallelTaskCount` (default 12) and returns results in input order. Options control the working and output digits, the two transport orders, the relative-error target, the singularity mode (`"Avoid"` or `"SingularityJump"`), and the message language; `"WorkDirectory"` denotes the temporary run root for bridge files. Errors surface as `Failure` objects with the bridge labels of section 5.1.

5.5 Options and common pitfalls

Working precision is a global setting: `OutputDigits` formats output but never raises internal precision. Executing a serialized plan above its planning precision is rejected (section 3.4). Exact routes reject `float`/`complex` input; numerical structure classification requires a declared reliable input precision. The batched singular-start entry accepts only paths whose transitions are all ordinary Taylor segments and writes no save files; individual calls to `transport_path_refined` cover those cases. Singularity-mode strings differ between the interfaces (`"avoid"`/`"singularity_jump"` in Python, `"Avoid"`/`"SingularityJump"` in Wolfram), and messages are English by default.

6 Examples

The package ships eight runnable examples. The numbers below were produced by fresh runs of the released scripts with `python-flint` 0.9.0 on Python 3.12 and Wolfram Language 15.0; the five examples that request output layouts leave JSON summaries under their own `results/` directories.

6.1 Scalar endpoints with save points

Three scalar examples exercise the singular-boundary records. The first, $\mathrm{d}y/\mathrm{d}x = 2y/x$ with exact solution $3x^2$, starts at the regular singular point $x = 0$ from the branch $\{a{=}2, b{=}0, C{=}[3]\}$ and transports to $x = 1$ at 50 working digits with primary and reference orders 20 and 28: the returned value is 3.00000000000000000000000000000000000000000000000000 with relative ball radius 7.7×10^{-57}, and the primary/reference relative difference is zero. The second, $\mathrm{d}y/\mathrm{d}x = y/x^2$ with solution $C\,e^{-1/x}$, starts at the decoupled irregular singularity $x = 0$ from the record $\{\phi{=}\{-1 \mapsto -1\}, a{=}0, b{=}0, C{=}[3]\}$: at $x = 1$ it returns $3e^{-1} = 1.10363832351432696490645972415842022278416101669833866 77$ with a spurious imaginary component of magnitude 7.9×10^{-57} (ball radius 3.5×10^{-114}), and the save file written at the singular start can be re-read by `exponential_boundary` to reconstruct the identical boundary object.

The third scalar example exercises multi-term exponential phases at a third-order pole. The diagonal system

$$A(z) = \operatorname{diag}\left(\frac{3}{2z^3} + \frac{1}{z^2} + \frac{1}{2z} + 2,\ \frac{1}{z^3} + \frac{1}{3z^2} + \frac{7}{2z} - 1\right) \tag{6.1}$$

has at $z = 0$ a decoupled essential singularity whose two sectors carry the polynomial phases $\Phi_1 = -3/(4z^2) - 1/z$ and $\Phi_2 = -1/(2z^2) - 1/(3z)$ and the residual powers $a = 1/2$ and $a = 7/2$; the boundary consists of two $\{\phi, a, b, C\}$ records, one per sector, each phase containing two negative powers in contrast to the single term above. The closed-form solution $(z^{1/2}e^{\Phi_1+2z},\ z^{7/2}e^{\Phi_2-z})$ equals $(e^{1/4},\ e^{-11/6})$ at $z = 1$; the transport with orders 48 and 64 and sampling fraction 0.45 reproduces these values with relative differences 6.0×10^{-25} and 3.9×10^{-25}, meeting the requested 10^{-14} target, and saves both endpoints.

6.2 Regulator Laurent-series reconstruction

The reconstruction example combines a zero connection (the boundary value is transported unchanged) with the symbolic boundary $1/\varepsilon + 2 + 3\varepsilon + 4\varepsilon^2 + 5\varepsilon^3 + 6\varepsilon^4$, so that the certified leading power -1 and every coefficient are known. The user supplies seven production candidates and two validation points, caps the first internal power at 2, and requests 12 digits. The first round consumes only the four-point prefix; its independent validation fails with relative residual 1.2×10^{-4}. The second round adds two candidates—reusing all four previous production solves and both validation solves—and passes with residual 5.7×10^{-44}, returning the pole coefficient 1 and the finite part 2 to better than 10^{-30}; one candidate remains unused. A second run with `sample_angle_range` $= (-\pi/3, \pi/3)$ places all automatic samples on exactly three interior rays inside the open sector and still certifies the digit goal.

6.3 Wolfram Language interface walkthrough

The interface example constructs the diagonal system $A(z) = \operatorname{diag}(1/(z-1), -2/(z+3))$ with $Y(0) = (1, 1)$, plans in `"Avoid"` mode at 80 working digits, and executes the

plan without replanning. Evaluated at the target $z = 1/2$, the primary/reference relative difference is 3.4×10^{-45}, the certification mode is reported as certified, and the final vector agrees with the closed form $(1/2,\, 36/49)$ through the first 39 digit at the requested 40 output digits.

6.4 Parallel batches of fixed regulator values

The equation $\mathrm{d}y/\mathrm{d}x = \varepsilon\, y/(1+x)$ with $y(0) = 1$ has the closed form $y(1) = 2^{\varepsilon}$. The Python entry evaluates $\varepsilon \in \{1/5, 1/6, 1/7\}$ in a pool requested at twelve workers (three effective) at 60 working digits with orders 112 and 128; for $\varepsilon = 1/6$ the value reproduces $2^{1/6}$ with relative difference 2.5×10^{-43} against the requested 10^{-40}, and both the internal error gate and the closed-form comparison pass for all three tasks. The Wolfram entry runs the same three jobs through `FlintNDEEvaluateEpBatch` with orders 48 and 64; the maximum deviation from 2^{ε} is 4.1×10^{-40} and all four consistency checks (including input-order preservation) pass.

6.5 Schwarzschild quasinormal modes

The final example is a two-ended radial connection problem from black-hole perturbation theory; it involves no integral family, and the transported pair is the reduced unified function and its derivative—the master function up to the known prefactor of equation (6.3).

Physical background. Gravitational perturbations of the Schwarzschild geometry separate, for each angular momentum ℓ, into odd- and even-parity sectors governed by the Regge–Wheeler and Zerilli master functions [13, 14]. At fixed (ℓ, ω) the even-parity function and its derivative are obtained from the odd-parity pair by the Chandrasekhar–Darboux transformation [35], so a single equation suffices. Quasinormal modes are the solutions that are purely ingoing at the horizon and purely outgoing at infinity; with these boundary conditions the frequency spectrum is discrete and complex [16, 17]. In units $2M = 1$, with $x = r/2M \in (1, \infty)$ and tortoise coordinate $r_* = x + \ln(x-1)$, the odd-parity function u_o^- obeys

$$\frac{\mathrm{d}^2 u_o^-}{\mathrm{d}x^2} + \frac{1}{x(x-1)}\frac{\mathrm{d}u_o^-}{\mathrm{d}x} + \left[\frac{\omega^2 x^2}{(x-1)^2} - \frac{\ell(\ell+1)x - 3}{x^2(x-1)}\right] u_o^- = 0, \tag{6.2}$$

with potential $V = (1 - 1/x)\,[\ell(\ell+1)/x^2 - 3/x^3]$.

From the second-order equation to the transported system. The prefactor

$$u_o^- = x^{\alpha_0}(x-1)^{\alpha_1 \mathrm{i}\omega}\, u, \qquad (\alpha_0, \alpha_1) = (3, -1), \tag{6.3}$$

removes the second-order finite poles: x^3 absorbs the algebraic behavior at the center $x = 0$, and $(x-1)^{-\mathrm{i}\omega}$ absorbs the ingoing plane wave $e^{-\mathrm{i}\omega r_*}$, since $r_* \sim \ln(x-1)$ at the horizon. With $w \equiv \mathrm{i}\omega$ entering exactly over $\mathbb{Q}(\mathrm{i})$ and $\lambda = (\ell-1)(\ell+2)/2$, the reduced function u satisfies $\mathrm{d}^2u/\mathrm{d}x^2 = -a(x)\,\mathrm{d}u/\mathrm{d}x - b(x)\,u$ with

$$a(x) = \frac{a_x}{x} + \frac{a_h}{x-1}, \qquad b(x) = -w^2 + \frac{b_h}{x-1} + \frac{b_x}{x}, \tag{6.4}$$

$$a_x = 5, \qquad a_h = 1 - 2w, \qquad b_h = 4 - 2\lambda - 2w^2 - 5w, \qquad b_x = -4 + 2\lambda + 5w, \tag{6.5}$$

and the transported pair is the companion vector $U = (u, u')^T$; the constant part of the companion matrix has eigenvalues $\pm w = \pm \mathrm{i}\omega$, the two asymptotic plane waves at infinity. Diagonalizing that constant part reorganizes the same system into simple poles $\Lambda_\infty + R_0/x + R_H/(x-1)$ with constant residue matrices; the example transports the companion form and lets the package's local machinery perform its own diagonalization at the endpoints.

Boundary data. Both endpoints are literal singular starts. At the horizon $x = 1$ the physical boundary condition—no outgoing flux—selects, in the reduced variable, the indicial remainder with leading power zero, entered as $\{a{=}0, b{=}0, C{=}[1, -b_h/a_h]\}$: after the plane wave is absorbed, the Frobenius series of the ingoing remainder starts at power zero, and the second component is fixed by the first-order correction $a_h c_1 + b_h c_0 = 0$. The outgoing remainder at the horizon is the forbidden branch $\{a{=}-a_h,\ b{=}0,\ C{=}[0,\ 1-a_h]\}$: u itself starts at power $1 - a_h$ while u' starts one power lower, so the record quotes the lowest common power $-a_h$. At infinity the system is inverted, $z = 1/x$, and the boundary records are

$$\begin{aligned} &\text{outgoing: } \{\phi{:}\, z^{-1} \mapsto w,\ a{=}3 - 2w,\ b{=}0,\ C{=}[1, w]\}, \\ &\text{ingoing: } \{\phi{:}\, z^{-1} \mapsto -w,\ a{=}3,\ b{=}0,\ C{=}[1, -w]\}, \end{aligned} \tag{6.6}$$

that is, $e^{\pm \mathrm{i}\omega r_*}$ times an algebraic remainder; the ingoing remainder carries z^3 from the x^3 prefactor, and the outgoing remainder the additional z^{-2w}, because $e^{+\mathrm{i}\omega r_*} = e^{\mathrm{i}\omega x}(x-1)^{\mathrm{i}\omega}$ contributes an extra power of x relative to the absorbed horizon factor. The vectors $C = [1, \pm w]$ are precisely the constant plane-wave eigenvectors. Zero leading powers therefore label the regular remainders of known asymptotic plane waves, not physical constant waves.

Computation. For the fundamental $\ell = 2$ mode the script fixes $w = 0.17792463137787093 + 0.74734336883608354\,\mathrm{i}$ over $\mathbb{Q}(\mathrm{i})$, corresponding to the tabulated $2M\omega = 0.74734 - 0.17792\,\mathrm{i}$ [16, 17]. It starts once at the literal horizon and once at the literal `inf`, transports the selected branches to a common matching point, and decomposes each vector in the opposite endpoint basis. Infinity is initialized by the formal asymptotic route of section 3.3: the leading matrix has simple distinct eigenvalues, the nearest root gap is $\Delta = 1.53646$, and the default rule $q = 3$ with requested order $N = 38$ places the first ordinary matching point at $x = 74.19640695257230$. Working at 70 digits with orders 38 and 55 and 88 and 120 Cauchy samples, the quasinormal-frequency block completes in 18.4 seconds and the shifted-frequency control in 20.0 seconds. The outgoing and incoming five-order block ratios at the first matching point are 2.90×10^{-3} and 3.20×10^{-3} with decreasing-block trend; the local vectors agree with an independently implemented scalar second-order recurrence to relative differences 3.3×10^{-36} and 9.4×10^{-36}, and at the fixed comparison point $x = 28$ to 6.0×10^{-19} and 2.5×10^{-18}. The maximum primary/reference transport difference is 3.75×10^{-22} against the requested 10^{-10}. The forbidden-to-allowed ratios are

Frequency	horizon $\to$ infinity	infinity $\to$ horizon
tabulated quasinormal frequency	2.4549×10^{-15}	3.7235×10^{-15}
relative shift 10^{-3}	3.9002×10^{-3}	5.9175×10^{-3}

The first row is consistent with zero at the demonstrated refinement scale, while the shifted row is the counterfactual mismatch expected when the boundary conditions are evaluated at a frequency that is not an eigenfrequency. The infinity expansion is a formal sectorial series and no convergence radius is claimed for it; this example tests a radial connection problem only.

Full radial profile at fixed accuracy. A third block keeps the horizon start alone and evaluates the complete radial profile of the mode at the same requested 10^{-10} relative accuracy. The sample points are chosen uniformly in the tortoise coordinate r_* on $[-6, 160]$—uniformly in x, the oscillation would compress infinitely toward the horizon—and inverted to $x > 1$ by Newton iteration; the grid additionally contains the exact point $x = 4$, whose value agrees with the independently planned horizon chain above to 5.2×10^{-32}. The 721 requested points are covered by 271 planned nodes at step fraction $\kappa = 0.06$: 585 of them are interior samples sharing their segment's Taylor coefficients, the largest bucket holding 32 points. The complete curve—seed transport from the horizon and both certification chains included—finishes in under one second, and each further requested point costs one dense evaluation rather than a new transport. Short steps are essential here: the profile grows like $e^{\mathrm{Re}\, w\, r_*}$ and reaches $|u_o^-| \simeq 2.8 \times 10^{12}$ at $r_* = 160$, so any truncation injected along the chain is amplified by the remaining growth. With the default planning cap $\kappa = 1/4$, which is ample near the $x = 4$ matching point of the two-ended computation, the two-chain comparison at the profile end fails outright; at $\kappa = 0.06$ the same orders 38 and 55 give an endpoint difference of 1.5×10^{-17}. Figure 3 shows the profile against r_* on the truncated range, with $\log_{10} |u_o^-|$ absorbing the twelve decades of magnitude: the oscillation damps through the potential-barrier region and grows exponentially toward both ends, the standard signature of a resonance eigenfunction that no finite norm can normalize.

7 Verification against independent references

Three layers of verification accompany the package. First, the shipped examples are self-checking: every script compares its output against a closed form, an independently implemented recurrence, or a counterfactual control before reporting success, as in section 6. Second, the Python and Wolfram test suites (181 unit tests and 26 end-to-end Wolfram checks) run the public interfaces against internal consistency requirements—round trips of every serialized format, rejection of malformed input, plan/execute separation, precision gates, save-point contracts and the exact-gate behavior of section 3.2—and were re-executed fresh for this paper. Third, independent validation directories exercise the released tree from empty result folders: a 16-dimensional exact system over $\mathbb{Q}(\mathrm{i})(k)$ cross-checks complete endpoint vectors between the package and a separately written driver, and the singular target classification is validated on translated targets (for example $2 + \mathrm{i}$ beside $5 + \mathrm{i}$) so that the

classification cannot degenerate to a special-cased letter or coordinate. The package does not ship a timing benchmark against other programs, and no relative performance claim is made here.

8 Summary and outlook

`FlintNDE` provides a fast, plain numerical solver for first-order matrix differential equations with rational or analytic coefficients, built on FLINT ball arithmetic so that every result carries a rigorous error estimate. Its certified workflow runs from exact structural input through generalized power series, adaptive path planning with plan/execute separation, and reusable save points to regulator-series reconstruction, and is exposed from both Python and Wolfram Language. The Schwarzschild quasinormal-mode example demonstrates that the same contract solves a two-ended radial connection problem with irregular endpoints, and the remaining examples cover each public interface.

The supported classes of irregular singularities are deliberately strict: the package certifies decoupled essential singularities and start-only second-order formal branches, and fails closed on ramified gauges, algebraic extensions beyond $\mathbb{Q}(\mathrm{i})$, general formal block decoupling and Stokes connections. Shipping public Feynman-integral examples with same-input benchmarks against existing programs, and promoting the regulator-neutral and epsilon-jet modules to top-level interfaces, are natural next steps.

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Statement on AI Use. The author designed the package notation, user-facing functions and input formats, and set the overall direction of the implementation. Several agent programs, each driven by a different large language model, implemented the package and its documentation under the author's guidance. The validation combines comparisons with independent references and cross-language consistency checks with internal certification gates.
The package is available at https://github.com/J7C/dSIBP-Packages.

A Public interface

The Python package exports its stable interface through `flintnde.__all__`; four modules (`regular_point_de`, `epsilon_jet_de`, `mathematica_bridge`, `savepoints`) are public at module level only. The Wolfram interface exports the symbols of section 5.4.

B Exact arithmetic, thresholds and serialized formats

The exact backend represents Gaussian rationals as pairs of FLINT `fmpq` and complex matrices as pairs of `fmpq_mat` of unchanged dimension, with square-free factorization, char-

Table 1. Public interface by layer.

Layer	Exported names
Numerics	`DEFAULT_WORKING_PRECISION_DIGITS`, `FLINTNDE_CITATION_NOTICE`, `configure_working_precision`, `acb_midpoint_matrix`, `column_vector`, `vector_norm_inf`, `relative_difference_inf`
Exact input	`GaussianRational`, `gaussian_rational`, `rational_function`
Systems	`RationalMatrixSystem`, `RationalFunction`, `analyze_singularities`, `SingularityInventory`, `SingularityRecord`, `AnalyticMatrixSystem`, `PartialFractionSystem`
Local bases	`RegularSingularSystem`, `build_frobenius_manifest`, `build_exact_frobenius_manifest`, `build_power_log_basis`, `PowerLogBasis`, `NumericalRegularSingularSystem`, `NumericalFrobeniusOptions`, `build_numerical_frobenius_manifest`, `build_local_solution_basis`, `LocalSolutionBasis`, `LocalReductionError`, `attempt_fuchsian_reduction`, `MoserBalanceTransformation`, `FuchsianReductionResult`
Boundaries	`frobenius_boundary`, `FrobeniusBoundary`, `FrobeniusBoundaryTerm`, `exponential_boundary`, `ExponentialBoundary`, `ExponentialBoundaryTerm`
Routing	`NamedPoint`, `classify_point`, `PointClassification`, `prepare_local_expansion`, `LocalExpansion`, `build_adaptive_path`, `build_adaptive_path_plan`, `PathPlan`, `AdaptivePath`, `AdaptivePathSingularityError`, `adaptive_path_to_json`, `adaptive_path_from_json`, `plan_transport_path`, `direct_user_point_path`, `transport_planned_path`, `transport_planned_path_refined`, `PlannedPath`, `SingularityJumpSpec`, `SingularityJumpBasis`, `SingularityJumpError`, `SingularPathError`, `planned_path_to_json`, `planned_path_from_json`
Singular targets	`plan_singular_target_match`, `evaluate_singular_target`, `SingularTargetResult`, `SingularTargetMatchPlan`
Transport	`build_straight_path`, `transport_path`, `transport_path_refined`, `transport_frobenius_boundaries_refined`
Reconstruction	`reconstruct_series_solution`, `SeriesReconstructionResult`, `SeriesValidationError`, `fit_sampled_series`
Parallelism	`run_ep_tasks`, `EpTaskBatchResult`, `DEFAULT_PARALLEL_TASK_COUNT`
Diagnostics	`five_term_tail_diagnostic`, `FiveTermTailDiagnostic`
Output	`initialize_output_layout`, `OutputLayout`
Wolfram	`$FlintNDEVersion`, `FlintNDERationalSystem`, `FlintNDEPartialFractionSystem`, `FlintNDEPlanPath`, `FlintNDEExecutePath`, `FlintNDEEvaluateEpBatch`, `FlintNDEBridgeError`

acteristic polynomials, nullspaces and reduced row echelon forms implemented over $\mathbb{Q}(\mathrm{i})$; no external computer algebra system is required for the certified routes. The numerical structure classification of section 3.2 uses the relative cutoff

$$\tau_{\mathrm{rel}} = N\,10^{-p}, \qquad \tau_{\mathrm{abs}} = \tau_{\mathrm{rel}} \max\left(\max_{i,j} |R_{ij}|, \max_{m,i,j} |(A_m)_{ij}| \right), \tag{B.1}$$

where N is the system dimension and p is the reliable decimal precision of the input, not of the arithmetic. The classification records its full provenance: input precision, both cutoffs,

the matrix scale, and the largest zeroed and smallest retained quantities, with undefined relative ratios reported as null rather than fabricated.

All serialized records—paths, plans, inventories, local manifests, save points and reconstruction results—carry versioned schema names (for example `flintnde_adaptive_path_plan_v1`, `flintnde_exponential_power_log_v1`, `flintnde_series_reconstruction_v1`) and are validated with a strict exact-keys gate: a record with a missing or unexpected field is rejected with a message that names the expected key set. Ball values serialize as midpoint, radius and exponent, and plans record their `planning_precision_digits`, which section 3.4 elevates to an execution gate. Results are laid out in fixed categories (`configuration`, `singularities`, `frobenius`, `transport`, `regularization`, `summary`) beside the calling script, never inside the package source tree.

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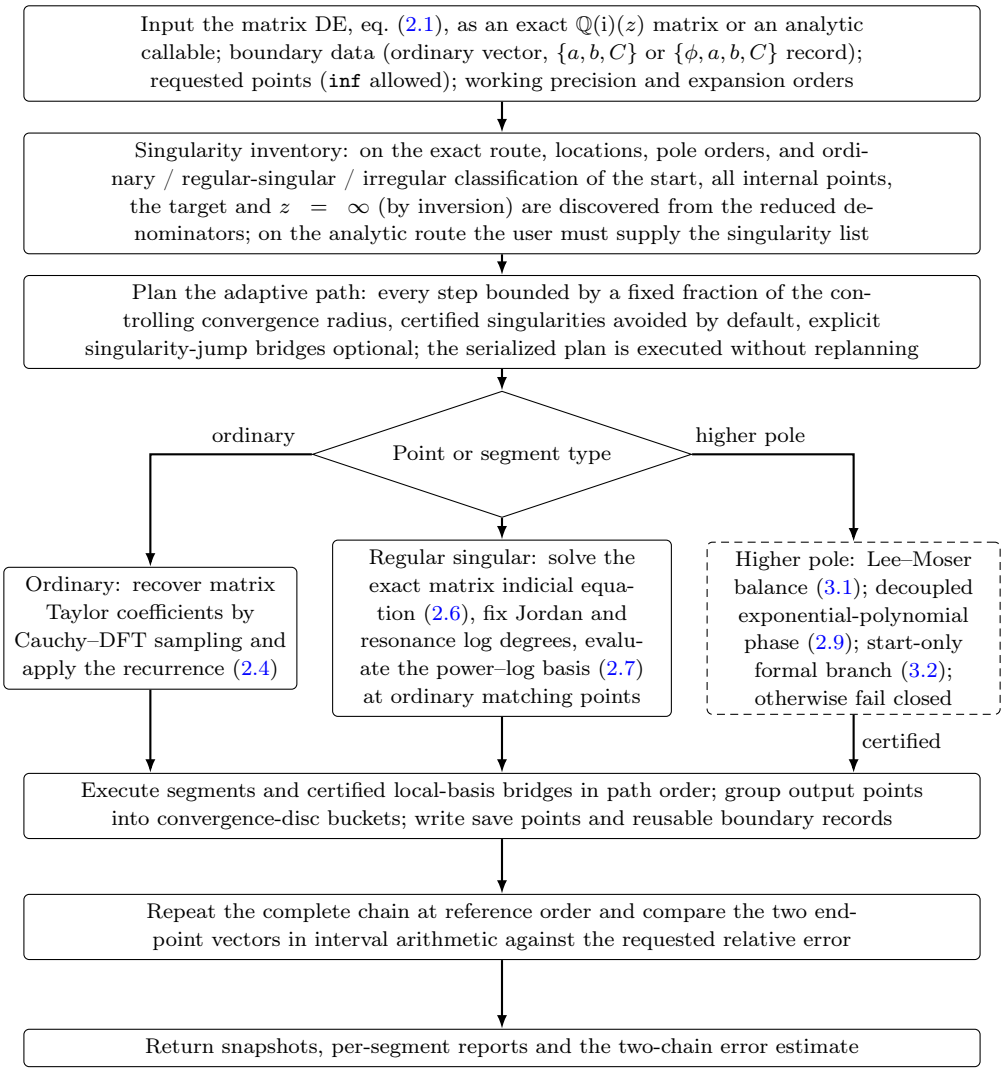


Figure 1. The numerical transport pipeline. Input formats and the scope contract are given in section 2; local solution structures in section 2.2; exact certification and the floating-point route in section 3.2; the three higher-pole routes and their fail-closed boundary in section 3.3; path planning, singularity jumps and save points in section 3.4; efficiency features in section 3.5; and two-chain error control in section 3.6.

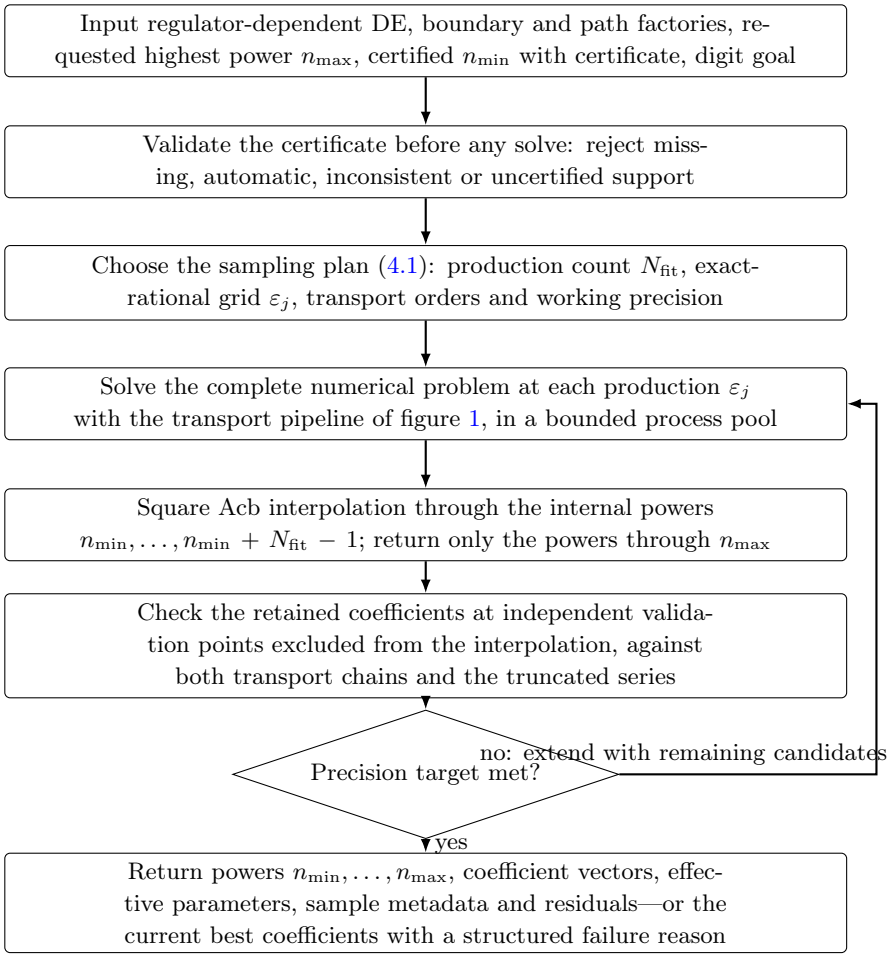


Figure 2. The regulator-series reconstruction workflow. Every production sample reuses the complete transport pipeline of figure 1; no separate solution-fitting engine exists. The sampling formulas are discussed in section 4.2 and the candidate-pool and validation semantics in section 4.3.

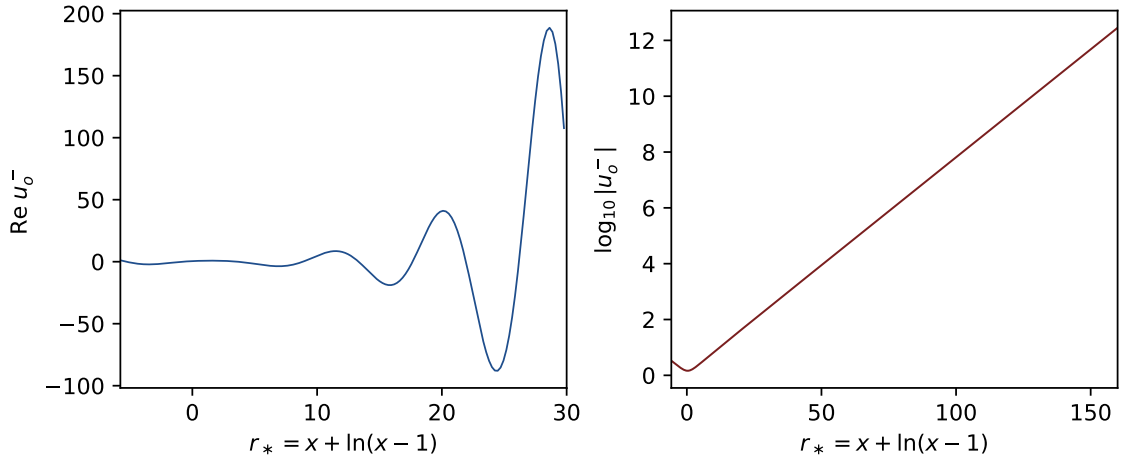


Figure 3. Radial profile of the fundamental $\ell = 2$ Schwarzschild quasinormal mode, normalized to unit ingoing amplitude at the horizon, evaluated by one planned transport with shared-coefficient buckets at requested relative accuracy 10^{-10}. Left: real part of the master function $u_o^-(x)$ in the near-barrier window $r_* \leq 30$, where it stays at natural scale. Right: $\log_{10}|u_o^-|$ across the full truncated domain; the magnitude spans twelve decades between the barrier region and the truncated outer end.